

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Valve Seat Puller Kit, Part Number 3822761
- Air compressor seat removal tool, Part Number 3377416
- Valve spring tester, Part Number 5299714
- Air compressor seat installation tool, Part Number 3377415

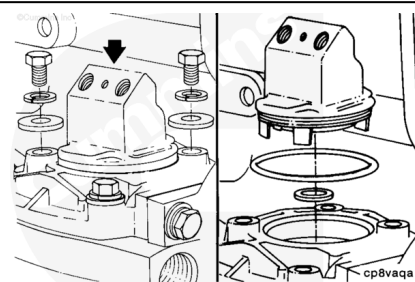
Additional Service Items

- Press
- Accrolube Lubrication Teflon™ Grease, or equivalent

Disassemble

⚠ WARNING ⚠

Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.



Hold the unloader valve body down. Remove the two captive washer capscrews and two plain washers.

Remove the unloader valve body.

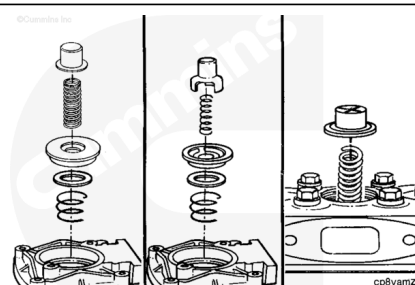
Remove the o-ring seal.

Remove the rectangular ring seal.

Remove the unloader valve cap and the unloader valve spring.

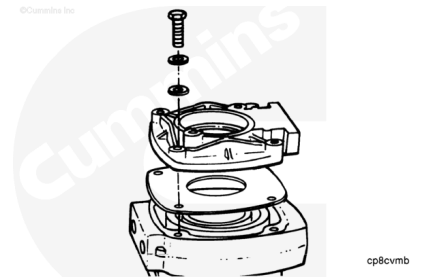
Remove the intake valve seat and valve.

Remove the intake valve spring.



Remove the capscrews, lock washers, and plain washers that hold the cover and head to the crankcase.

Remove the cover and discard the gasket.



Remove the head and discard the gasket.

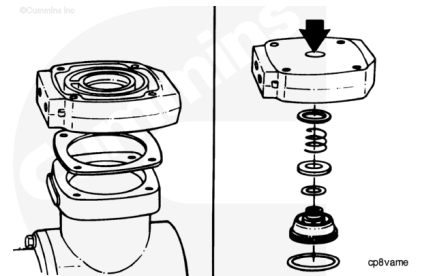
Hold the head bottom side up. Use thumb pressure to remove the exhaust valve seat assembly.

Note : If the exhaust valve seat assembly can not be removed by thumb pressure, use valve seat puller, Part Number 3822674. A press and air compressor seat removal tool, Part Number 3377416, can be used to remove the exhaust valve seat.

Remove the o-ring seal from the exhaust valve seat or head.

Remove the exhaust valve.

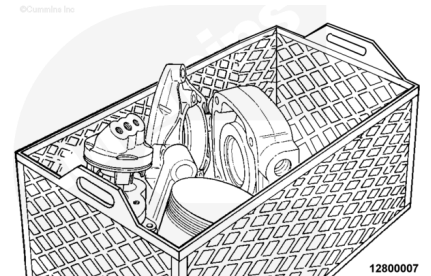
Remove the compression spring and wear plate from the head.



Clean

⚠ WARNING ⚠

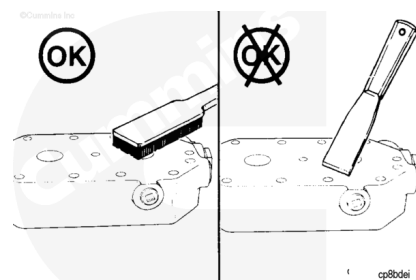
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Soak the parts in a kerosene emulsion-based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier **must** be contacted about solution concentration, temperature, and soak time.

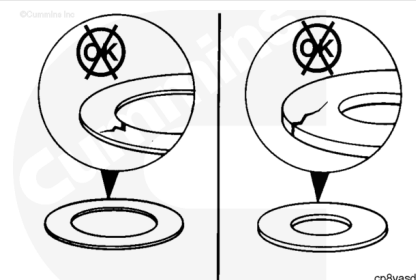
Note : Do not use a scraper to remove carbon and scale, the sealing surfaces can be damaged.

The parts can be scrubbed with a stiff nonmetallic bristle brush.



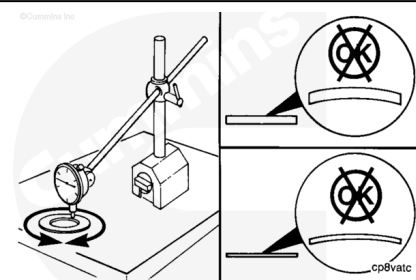
Inspect for Reuse

Inspect the intake and the exhaust valves for cracks or other damage.



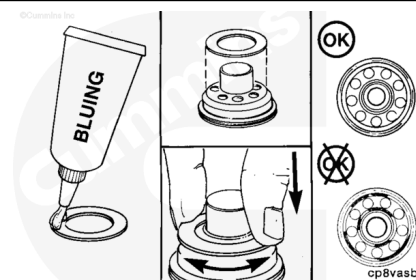
Measure the flatness of the intake and exhaust valves. Both valves **must** be flat within 0.03 mm [0.001 in].

Note : Holset Engineering Co., Inc. recommends new valves be installed.



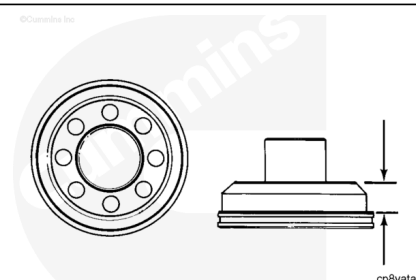
Apply a bluing compound to the exhaust valve seating surface.

Install the exhaust valve on the valve seating surface and check the seating area. Replace the seat if the contact area is **not** 100 percent.



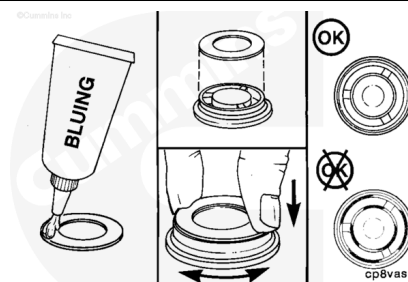
Measure the exhaust valve seat height.

If the seat height is less than 12.32 mm [0.485 in], replace the seat.



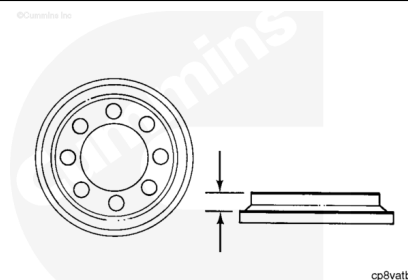
Apply a bluing compound to the intake valve seating surfaces.

Install the intake valve on the valve seating surface and check the seating area. Replace the seat if the contact area is **not** 100 percent.



Measure the intake valve seat height.

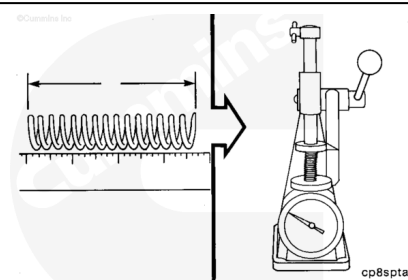
If the seat height is less than 6.86 mm [0.270 in], replace the seat.



Use valve spring tester, Part Number 5299714, to check the intake, exhaust, and unloader valve springs. Replace any springs that do **not** meet specifications.

Note : Holset Engineering Co., Inc. recommends new springs be installed.

See the Compressor Spring Force Specifications chart shown below.



Load Required to Compress Spring to Measurement Length

Spring Part Number	Spring Use	Approx. Free Length: mm [in]	Number of Coils	Wire Diameter : mm [in]	Measure ment Length: mm [in]	Minimu m: kg [lb]	Maximu m: kg [lb]
128080	Exhaust valve	17.02 [0.670]	3.0	2.03 [0.080]	7.11 [0.280]	3.6 [8.55]	4.7 [10.35]
190334	Intake valve	12.70 [0.500]	2.8	1.58 [0.062]	7.11 [0.280]	0.35 [0.65]	0.5 [1.10]
150631	Unloadin g valve (naturally aspirate d) center unloadin g valve - twin	41.91 [1.650]	11.5	2.03 [0.080]	24.89 [0.980]	14.5 [32.00]	17.2 [38.00]

Load Required to Compress Spring to Measurement Length							
3023101	Unloading valve (all turbocharged engines)	41.91 [1.650]	10.8	1.65 [0.065]	24.89 [0.980]	5.9 [13.00]	7.7 [17.00]
3049553	E-Type unloader valve	41.91 [1.650]	11.25	1.93 [0.076]	24.89 [0.980]	10.4 [23.00]	12.2 [27.00]
800399-XW	Unloader	17.02 [0.67]	6	1.04 [0.041]	6.60 [0.260]	N/A	2.54 [5.60]
					10.03 [3.95]	1.71 [3.78]	N/A
802000-FZ	Intake and exhaust	10.16 [0.40]	4.25	0.79 [0.031]	4.57 [0.18]	N/A	0.36 [0.80]
					5.99 [0.275]	0.20 [0.45]	N/A
3054489	Exhaust valve	21.49 [0.846]	4.5	2.54 [0.100]	15.21 [0.599]	3.88 [8.55]	4.74 [10.45]

Assemble

Install the wear plate and compression spring in the head as shown.

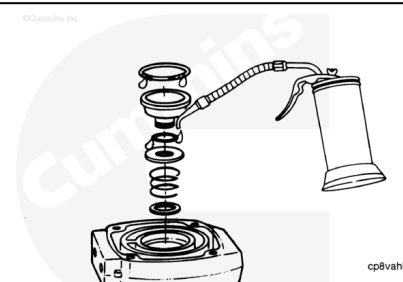
Install the exhaust valve on the exhaust valve seat.

Install the o-ring seal on the seat.

Use clean 15W-40 oil to lubricate the o-ring seals.

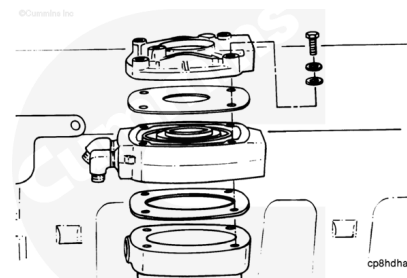
Use hand pressure to install the exhaust valve assembly in the head.

Note : For SS-296 and ST-676 air compressor models, a press and air compressor seat installation tool, Part Number 3377415, can be used. Exhaust valve seat puller, Part Number 3822674, can also be used to install the valve seat found in the Valve Seat Puller Kit, Part Number 3822761.



Use new gaskets to install the cylinder head and cover.

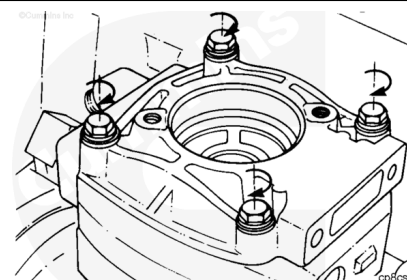
Install the flat washers, lock washers, and capscrews.



Tighten the cylinder head capscrews of the single cylinder compressor in an alternating sequence.

Torque Value:

1. 7 n•m [62 in-lb]
2. 14 n•m [124 in-lb]
3. 20 n•m [177 in-lb]
4. 27 n•m [239 in-lb]



Install the intake valve spring with the tang down.

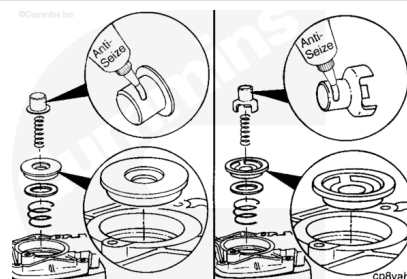
Install the intake valve.

Install the intake valve seat with the flange side up.

Install the unloader valve cap spring.

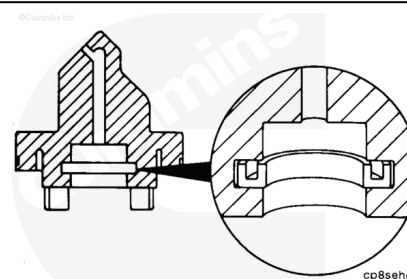
Install the unloader valve cap.

Use high temperature grease (Accrolube Lubrication Teflon™ Grease, or equivalent) to lubricate the outside diameter of the cap.

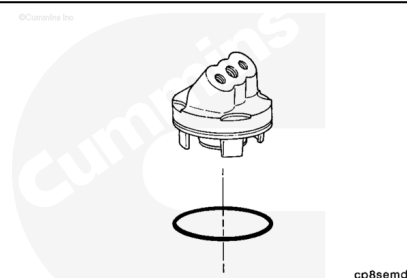


Install the new rectangular ring seal into the unloader body.

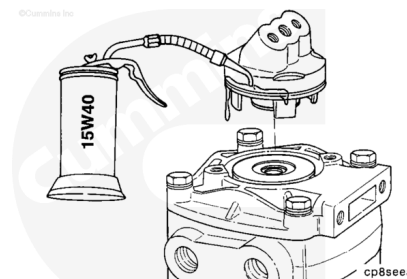
Note : The seal must be installed with the grooved side up.



Install a new o-ring seal on the unloading valve body.



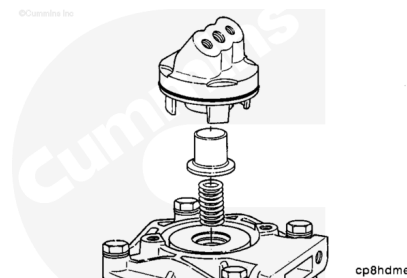
Use clean 15W-40 oil to lubricate the seal.



Install the compression spring.

Use Accrolube Lubrication Teflon™ Grease to lubricate the outside diameter of the unloader cap and install it into the unloading valve body.

Install the unloading valve body.



⚠ CAUTION ⚠

Do not overtighten these screws. Compressor damage will result.

Hold the unloader body down. Install the two plain washers and captive washer capscrews.

Tighten the capscrews.

Torque Value: 14 n•m [124 in-lb]

